Meng Wu

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Education

University of Michigan Ann Arbor, MI
M.S. Electrical & Computer Engineering — Signal, Image Processing & ML; **GPA 4.00/4.00**
Sep 2024 – Jun 2026

Ph.D (Incoming 2026 Fall at University of Michigan. Advisor: Qing Qu)
University of Electronic Science and Technology of China (UESTC) Chengdu, CN
B.E. Electronic Information Engineering; **GPA 3.76/4.00 (Major 3.98, Rank 2/42)**
Sep 2020 – Jun 2024

Publications

Generalization of Diffusion Models Arises with a Balanced Representation Space
Zekai Zhang, Xiao Li, Xiang Li, Lianghe Shi, **Meng Wu**, Molei Tao, Qing Qu. *ICLR 2026*

Coarse-to-Fine Hierarchical Alignment for UAV-based Human Detection using Diffusion Models
Meng Wu*, Wenda Li*, Lianghe Shi, Sungmin Eum, Heesung Kwon, Qing Qu. *ICRA 2026, Under Review*

A Closer Look at Model Collapse: From a Generation to Memorization Perspective.
Lianghe Shi*, **Meng Wu***, Huijie Zhang, Zekai Zhang, Molei Tao, Qing Qu. *NeurIPS 2025,* ***(Spotlight, Top 3%)*** .

VIP: Versatile Image Outpainting Empowered by Multimodal Large Language Model.
Jinze Yang, Haoran Wang, Zining Zhu, Chenglong Liu, **Meng Wu**, Mingming Sun. *ACCV*, 2024.

Convolution Meets Transformer: Efficient Hybrid Transformer for Semantic Segmentation with Very High Resolution Imagery.
Yuji Wang, Ruojun Zhao, Shicai Wei, **Meng Wu**, Luo Yang, Chunbo Luo. *IEEE IGARSS*, 2024.
(* means equal contribution).

Research & Intern Experience

Graduate Research Assistant (Paid) Sep 2024 – Present
University of Michigan, Ann Arbor | Advisor: Prof. Qing Qu

1. Model Collapse and Generalization Study(*NeurIPS 2025* ***Spotlight, Top 3%***)

- Empirically analyzed model collapse from the perspective of generalization degradation in diffusion models.
- Explained the theoretical connection between model generalization and the entropy of training data distributions.
- Developed entropy-boosted data selection methods and curriculum priors that mitigate collapse while preserving sample fidelity.

2. UAV Object Detection via Diffusion-based Domain Adaptation (ICRA 2026 submission)

- Proposed a hierarchical Sim2Real diffusion framework (HSRA) to align synthetic and real UAV imagery.
- Designed a two-stage coarse-to-fine noise alignment mechanism improving domain adaptation robustness.
- Achieved +7.1% mAP on the VisDrone benchmark compared to standard YOLOv8 baselines.

AIGC Research Intern Apr 2024 – Jul 2024
Shengshu AI, Beijing, CN

- Investigate fixed-identity image and video generation with diffusion models and temporal attention alignment.
- Proposed diffusion-representation alignment improves CLIP similarity by 6% and reducing FID by 8.4 on internal benchmark.

Research Intern Nov 2023 – Apr 2024
Baidu Research, Beijing, CN

- Lead video editing project using Diffusion Transformer; fine-tune U-ViT with multimodal conditioning.
- Build 120k-poster dataset and fine-tune LLaVA via DPO, decreasing hallucination by 12%.

- Conduct experiments for an image-outpainting study accepted by ACCV 2024.

Teaching Assistant — Course 3036 AI & ML Aug 2023 – Feb 2024
UESTC, Chengdu, CN

- Lead weekly labs; design homework; mentor 90 students.

Projects

Efficient Text-to-Image Fine-Tuning (PyTorch, Diffusers, LoRA) Jul – Dec 2023

- Integrate Textual Inversion, LoRA, and Custom Diffusion into UniDiffuser for rapid domain adaptation.
- Design cross-attention adapter reducing fine-tuning time to ∼30 min on a single A6000.
- Achieve SOTA CLIPScore on T2I-CompBench; project earned 3rd Prize in Greater Bay Area Algorithm Competition.

Interactive Image Editing via LLM Guidance (Stable Diffusion, LLaMA-2, Gradio) Jun 2023 – Jun 2024

- Implement classifier-free guidance for automatic region selection, enabling conversational image editing.
- Release open-source demo (4k+ GitHub stars) integrating SD XL with a conversational front-end.

Honors & Awards

- The 3rd Prize, Greater Bay Area (Huangpu) International Algorithm Competition (Dec 2023)
- UESTC Academic Merit Scholarship (Oct 2023)
- UESTC Most Contributive Student Scholarship (Sep 2023)
- UESTC Academic Merit Scholarship (Nov 2022)
- The 3rd Prize, 2022 National College Student Embedded Systems Design Contest (Sep 2022)
- UESTC Academic Merit Scholarship (Nov 2021)

Technical Skills

Languages:	Python, C/C++, MATLAB
Frameworks / Libs:	PyTorch, Diffusers, HuggingFace, OpenCV, scikit-learn
Dev Tools:	Git, Docker, SLURM, Bash